Preparations for external use

The skin can easily absorb remedies through hair follicles and into the bloodstream, making it a convenient pathway for treatment.

Herbal Baths

A warm bath infused with fragrant herbs can be a relaxing experience and is especially helpful for treating children and babies. You can fill a muslin bag with fresh or dried herbs and place it under the tap. Strong herb infusions can also be added to the bath water, and you can soak in it for about 15-20 minutes. Hand and foot baths are a great way for pregnant women to relieve some of the swelling from the extra body weight.

Salves and Creams

Salves are a fun and effective way to prepare remedies. To make a salve or cream, you need 1 cup of infused herbal oil to 1 oz of beeswax. The key is making the infused oil, which can be done through two methods: heat processed or cold processed. The heat process takes only 10 minutes, while the cold process takes about 2 weeks. To make the salve or cream, place 1 oz of beeswax into a double boiler to allow melting, then pour in 1 cup of infused oil. Heat slightly until the oil and beeswax are mixed, then pour into prepared containers. Allow it to cool, label, and store in a dark place. You can also add about 50 drops or 1 tsp of essential oil to the salve before pouring it into the containers.

Ointments

To prepare an ointment, melt the beeswax in a double boiler, add 1 tsp of cocoa or shea butter or vegetable glycerin, add the herb powder, and leave it on heat for 20 minutes (some say 2 hours) until the herbs become infused. Transfer the mixture into a jelly bag or cheesecloth tightened with a rubber band or string and pour the mixture till the rim of the jar. Be sure to wear rubber gloves when working with this mixture and pour it into the jar while it is warm, as it can solidify as soon as it starts cooling. I do not recommend using paraffin wax or petroleum jelly to make any remedy, as they are processed and contain synthetic hardening agents.

Poultices

A poultice is a dampened mixture applied to the part of the body that needs the remedy. Fresh or dried herbs can be used as a poultice, and they are placed between two pieces of gauze or a cabbage leaf. If using fresh leaves, stems, or roots, break or crush them before utilizing them. If using dried herbs, a little warm water or jojoba oil can be added to the finely chopped or powdered herbs. After the mixture is ready to use, spread the paste into the gauze or cabbage leaf. Bind the gauze poultice to the affected area with a light cotton bandage, and keep it warm with a hot water bottle, warm towel, or warm rice sachet. An infused poultice can also be made by making a hot processed infusion.

Herbal, Aromatic, and Fragrance Oils

Essential oils are products of aromatic plants, extracted through a method called steam distillation. They can be purchased from health food shops. To make 'herbal oils' at home, infuse finely chopped herbs into pure vegetable oils such as almond, sunflower, or olive oil. This process can take approximately two weeks or longer and requires a large amount of herbs. Place the herbs and oil into a tight-fitting glass jar, keep it covered, and place it by the windowsill where sunlight can come in. Shake it every day. After two weeks, once the oil has absorbed the constituents, filter the mixture through a muslin cloth and store it in an airtight dark colored bottle. A tool called the Flora Lab can also be purchased to make herbal oils from Herbal Blessings.

Infused Oils

Massage oils, ointments, and creams consist of externally used active plant ingredients. Infused oils can be preserved up to one year if kept properly in a cool and dark place, but making small and fresh quantities of infused oils is more effective. There are two types of techniques to make infused oils. The hot process is best suited for comfrey, chickweed, or rosemary. The cold process is most appropriate for marigold and St. John's wort. The process for the cold-infused oil can be repeated by using new herbs, and the oil infused once can be set aside for storage for a few weeks.

Preparations from Fresh Plant Material

Plant Juice – Definition: The liquid obtained by pressing and crushing freshly harvested plant material (plant parts). – Preparation: Prepared from freshly harvested plant parts, expressed juice primarily contains water-soluble plant constituents. – Storage: Once opened, the bottle should be closed and stored in a refrigerator. The contents should be discarded after one week, or after the expiration date specified on the label. – Medicinal Action: Usually relatively weak, except in rare cases (expressed Echinacea juice, for example).

Distillates – Definition: Formulations obtained by extracting active principles from fresh or dried plant material by steam distillation. – Preparation: Obtained by separating the steam-volatile constituents of fresh or dried medicinal plants by vaporization. – Storage: See p. 5. – Medicinal Action: Determined by the water-volatile constituents (e.g., mustard oils, essential oils) contained in the distillate.

Oily Extracts – Definition: The preparation obtained by dissolving the fat-soluble constituents of a medicinal plant in, for example, olive oil, almond oil, or peanut oil. – Preparation: Prepared by immersing the freshly cut or dried plant parts in a vegetable oil and allowing them to stand, usually at room temperature, until the fat-soluble constituents have been extracted.

Storage: These preparations are relatively unstable and should therefore be prepared in small quantities (see p. 5 for storage instructions).

Extracts

Definition: Extracts are prepared by dissolving medicinal plants in a solvent to separate their active principles from extraneous substances. The type of formulation (aqueous, alcoholic) depends on the type of solvent used (water, alcohol). Extracts are characterized as either dry or liquid depending on the concentration of residual solvent in the final product. Note: The composition of and uses for preparations from the same plant may differ between the various preparation techniques. Hence, any extract produced by a special preparation technique is a unique active substance.

Aqueous Extract – Definition: An extract prepared using water as the extracting agent. These extracts mainly contain water-soluble constituents and few lipid-soluble components. – Disadvantages: The individual constituents are relatively unstable, and microorganisms can multiply rapidly.

Alcohol Extract – Definition: An extract prepared using ethanol and water mixed at various ratios and concentrations, called the menstruum. – Preparation: In Germany, the standard procedure calls for the plant material to be cut into pieces and then steeped in the menstruum (macerated) for several hours; in the North American standard, the fresh or dried and powdered plant material is macerated in the menstruum for a week to 10 days and the liquid is then pressed out with a hydraulic press (if available).

The drug is repeatedly steeped in the menstruum, strained and concentrated (percolation) until completely extracted. – Advantages: Ethanol (grain alcohol) is an excellent extracting agent. At higher volume concentrations, it can also extract lipophilic drug constituents, such as essential oils. Alcohol preserves the extracts for a longer time (up to 2 to 3 years) and assists rapid absorption of the active ingredients.

Tincture – Definition: A solution prepared by macerating or percolating a medicinal herb in various concentrations of ethanol. According to the German Pharmacopeia (DAB 1996), dry extracts made by using suitable concentrations of ethanol are also defined as tinctures.

Preparation: Tinctures are made by mixing 1 part drug with 5 parts extractant. However, for high-potency constituents like atropine from belladonna, the ratio is 1 part drug to 10 parts extractant. The process of extracting the active ingredients from herbs and plants can be a complex and nuanced process, but it is important to understand the different types of extracts available and how they are made.

Labeling: A dried herb to extract ratio (HER) of 1:5 or 1:10 means that the preparation was made using 1 part plant material and 5 or 10 parts of the menstruum. This ratio can help consumers understand the potency of the extract and how it was prepared.

Storage (see p. 5): Tinctures should not be stored for more than one year because some compounds in the extract may become unstable. Bauer and his co-workers found that echinacea tincture is stable for 2 to 3 years under normal conditions, with about 30% reduction in some important constituents per year (Personal communication, Rudi Bauer, Ph.D., 1987). It is important to carefully monitor the shelf life of tinctures and other extracts to ensure that they are still effective.

Note: Many tinctures should not be used undiluted. It is important to follow the instructions provided by the manufacturer and consult with a healthcare professional before using any herbal supplements.

Fluid extract (fluid extract) Definition: An alcoholic preparation of a medicinal herb containing higher concentrations of plant constituents than are found in conventional tinctures. One part of fluid extract corresponds to one part of the parent herb calculated on a dry-weight basis. This concentration is achieved by evaporating off some of the alcohol. Fluid extracts are a powerful and concentrated form of herbal medicine that should be used with caution.

Preparation: Fluid extracts are prepared in the same way as tinctures, but involve additional steps to increase the concentration of active ingredients. The process can be complex and may involve specialized equipment and techniques.

Dry extracts (powdered extracts),Definition: Solid preparations obtained by condensing and drying fluid extracts. A powdered extract generally contains 95% solids and 5% water residue (moisture). A native extract or native dry extract contains only plant extract material and is free of additives. Powdered extracts are a convenient and easy-to-use form of herbal medicine that can be added to food or drinks.

Preparation: Powdered extracts are made by carefully drying and condensing fluid extracts to create a concentrated powder. Adjuvants and carriers such as highly dispersed silica, lactose, and methylcellulose are sometimes added to prevent caking and to adjust the final extract concentration. Extracts that are adjusted to 5:1 (meaning that 1 part of the finished extract is equivalent to all the desirable and active constituents of 5 parts of the dried herb) are typical.

The package labeling provides information on the parent plant or plant part, the extractant, and the dried herb/extract ratio (HER). An HER of 10:1 means that 10 parts of the plant material yielded 1 part extract. A low HER (such as 1:2) indicates a high concentration of the active compounds in the plant material.

Special extracts Definition: Extracts that undergo special extraction and purification processes to separate, concentrate, and free them from toxic and undesirable substances. Special extracts are a highly concentrated and potent form of herbal medicine that should only be used under the guidance of a healthcare professional.

Preparation: Special extracts are made from raw extracts and undergo additional extraction and purification processes to increase the concentration of desirable active principles. Toxins and undesirable substances that do not contribute to the medicinal action of the preparation are removed. This treatment reduces the risk factors associated with the native material, and often provides a finished product that is many times more concentrated in one or more active constituents than is the parent herb, such as the highly concentrated and standardized ginkgo extract typically sold in consumer products (50:1). Hence, the effects of the special extract can no longer be compared with those of the raw materials. It is important to carefully consider the potential benefits and risks of using special extracts and to consult with a healthcare professional before use.

Medicinal teas are a type of herbal tea that are known for their strong medicinal properties. While some milder teas, such as ginger, peppermint, and chamomile, are considered safe to use at home, it is best to use medicinal teas under the guidance of an experienced healthcare professional like a naturopathic practitioner or herbalist. The concentrations of active principles in medicinal teas are dependent on several factors, including the proportion of herbs to extractant (water), the degree of cutting or grinding of the herbs, the recommended water temperature, and the steeping time.

Despite their many benefits, there are some disadvantages to using medicinal teas. It can be difficult to determine the exact dosage, and combined components may become separated. Additionally, cutting or powdering of herbs can often drastically reduce their shelf life by allowing oxygen to reach and degrade active constituents.

If you're interested in trying medicinal teas, there are many suppliers to choose from, including pharmacies, drug stores, health food stores, herb shops, and supermarkets. When preparing medicinal teas, it is important to cut leaves, flowers, and other plant materials coarsely to finely, while wood, bark, and root materials should be cut finely or pulverized. To ensure that your medicinal teas retain their potency, it is important to store them in metal tins or dark jars to prevent direct light exposure. The expiration date should be indicated on the label.

There are several types of tea preparations to choose from, including loose tea, tea bags, and instant teas. Loose tea is available as single-herb preparations and as tea mixtures, while tea bags are small filter bags in which finely chopped herbal teas are enclosed. Instant teas are tea preparations that readily dissolve in water, and are available in both powdered and granular forms.

When preparing your medicinal teas, there are several methods to choose from, including infusion, decoction, and maceration.

Infusion is a preparation made by pouring boiling water onto the fresh or dried herbs and allowing them to steep to extract their medicinal principles. This method is suitable for the preparation of delicate or finely chopped herbs with volatile and thermolabile constituents. Decoction, on the other hand, is prepared by boiling fresh or dried herbs in water for 10 to 60 minutes to extract their medicinal principles.

This method is suitable for the preparation of hard or very hard plant materials or herbs with sparingly soluble constituents. Finally, maceration, or cold extract, is prepared by allowing a tea herb to steep in cold water for several hours to extract its active principles. This method is suitable for the preparation of mucilage-containing herbs such as flaxseeds or psyllium seeds whose high concentrations of starches and pectins would cause them to gelatinize if prepared with boiling water. It is also used to prevent the extraction of undesirable constituents that dissolve in hot water.

MAKING DIFFERENT FORMS OF HERB

When creating different herbal forms such as lotions, shampoos, soaps, etc., it is important to research the herbs included in your recipe. Keep in mind that not all recipes found in books or on the internet are accurate. Try to avoid using herbs that have the same properties in your remedies, as some herbs blend well with others while others do not.

When blending your herbs, be aware that your first attempt may not turn out as expected. Don't worry, this is normal and practice makes perfect. The more you research and test your remedies, the better you will become. My family and I played a significant role in volunteering to be my "guinea pigs" in testing my remedies, and my co-workers also helped by trying them out to help me perfect my recipes. It took a lot of research, time, effort, and failures to get the perfect remedy, but in the end, I had a stack of papers with provings and a great recipe to help others. Provings are case studies and field studies done on a recipe.

To gather data, I would ask people to volunteer to use an herbal product and sign a waiver stating that it was a research project, and I wouldn't be held liable, etc. I would then have them write down how they felt, their emotional state, and how they liked the recipe in a wellness check journal that I provided. This journal helped me understand the potency of the recipe. For instance, my uncle suffers from diabetes and high blood sugar, and I have found a veggie capsule made up of a blend of herbs that he has been trying for one week. Already, he has cut back on his insulin medication while carefully monitoring his blood sugar every two hours. We have taken all safety measures and precautions in this case.

Tip: It is illegal to prescribe medicine in the state of Kentucky without a license. However, some herbal supplements are not approved by the FDA as medicine. You can consult someone in herbal supplements, but it is their own free will if they use the remedies. If you plan on using herbal medicines to help people, suggest rather than prescribe any remedy.

Before we begin to get our supplies to make an herbal remedy we must use all safety and health precautions that we can.

- Always wear gloves
- Protect the work surface
- Use special pots to cook with, not the ones you cook food in
- Use stainless steel, carbon steel, or glass pots and containers and utensils
- Always wash each item before transferring to a new remedy
- Make sure all containers, utensils, work surface is clean and dry before use
- Wear an apron to protect clothing
- Do not use metal tea balls or spoons, the herbal oils can interact causing a chemical reaction
- Warning some pots, containers may be stained with the herbal dye.
- Follow all Food Safety, GSM, FDA & USDA Guidelines with ingredients, labeling, dispensing, batch record, and so on.

For making the different forms of herbs you will need some supplies; See additional PDF.

- Medium size cooking pot
- Small size cooking pot
- Wooden spoons
- Strainer Plastic
- Double boiler
- Quart and pint glass jar with lids
- Muslin or cheesecloth
- Rubber bands
- Measuring spoons and cups plastic or wooden
- Distilled water
- Colored glass jars with droppers for oils
- Glass Jars with lids 2oz or 4oz
- Plastic Cosmetic Grade small containers with lids
- Whisk
- Address labels
- Permanent marker
- Cloth Tea Bags
- Wooden Stir Sticks
- Wooden or Ceramic Mortar and Pestle
- Hot Plate
- Vegetable Parchment Paper
- Ceramic Tile
- Wooden Craft Sticks
- Brown Paper Bags
- Syringe or pipette

TOPICS TO REMEMBER

Remember these important topics for creating safe and effective herbal remedies:

- Always wear protective clothing, including gloves. Lotions, medicines, nail polish, and even nicotine can contaminate the remedy.
- Use separate spoons or wooden stir sticks for each ingredient or combination of ingredients. Clean each stick with white vinegar after use.
- Have designated herb cooking supplies, such as pots, spoons, measuring cups, and measuring spoons, that are not used for everyday cooking to avoid cross-contamination.
- Keep your cooking area clean and free of food particles.
- Use ceramic hot plates to cool remedies.
- Have flour or cornmeal on hand in case of spills.
- Do not make remedies for others while sick. Have an understudy make the remedy instead.
- Use clean pots, pans, and stir sticks, and clean them with white vinegar (not bleach) or a mixture of lemon juice and baking soda. Use a clean brush, sponge, or loofah to clean herb pots and paper towel or muslin cloth to dry.
- Store your herbal kitchen supplies separately from everyday cooking supplies.
- Use glass or culinary/cosmetic grade remedy containers with labeled dates. Avoid using tin containers as some herbs interact with them. If using glass, use plastic seal lids and protect the jar rings with muslin cloth to avoid mold and rust.
- Never heat herbal remedies in plastic measuring cups. Use glass instead as formaldehyde can interact with the remedy.
- While it is not recommended, microwaves can be used to create herbal remedies in a pinch. However, gas stoves, wood burning stoves, and portable camping stoves can contaminate the remedy. The best option is an electric stove or hot plate, although some herbalists believe that no heating of remedies is safe.
- Keep a remedy making journal and back it up electronically and in writing to avoid losing important formulas.
- Safety is paramount when creating and using herbal remedies. Know your herbs and how to use them correctly and safely to protect your reputation and produce effective remedies.
- Keep your recipes safe and put a copyright logo and date on them to prevent theft. However, it is acceptable to alter a recipe to fit specific needs.
- Sanitization and safety are critical when dealing with herbs, both in purchasing and preparing them. Importance should be placed on preparing the food or sanitizing medical equipment to avoid harmful bacteria.

- Even experienced herbalists may not realize the importance of safety and cleanliness. Nicotine on hands, lotions, and other substances can contaminate remedies. Use separate utensils and follow guidelines to avoid reactions between herbs.
- With cooking with herbs & spices please remember food safety & regulations.

Practice Safe Food Handling-

WASH YOUR HANDS!

To maintain good hygiene, follow these guidelines:

- Wash your hands often, especially after touching food, handling garbage, smoking, using the restroom, or handling chemicals/cleaner.
- Keep your hands away from your face and hair to prevent the transfer of germs from the nose to the mouth.
- Keep all areas clean, including the stockroom, freezer, and fridge.
- Use separate equipment for herbs.
- Keep garbage covered and away from the remedy making area.
- Store herbs/foods six inches above the floor in a cool, dry area away from sunlight.
- Seal everything and label it with the expiration date and/or creation date. Make sure all ingredients are listed in order from greatest to smallest.
- List the potency as 1X, 2X, and/or 3X.
- Place warning labels on items when necessary.
- Cross Contamination-

Cross contamination can occur in three ways:

1. People to food
2. Food to food
3. Equipment to food

To Prevent Contamination-

- Wear gloves.
- Use sanitized tools.
- Sanitize the workspace.
- Keep all uncooked foods away from cooked foods.
- Have separate herbal cooking equipment from everyday cooking equipment.

Studying each individual herb can take a lifetime. There are so many herbs out there. You can use the common backyard herbs that you can wild harvest, or you can purchase them from a company, co-op, or farmer's market. The choice of herbs is entirely up to the herbalist. Studying an herb may only come to some for specific health reasons and/or symptoms rather than a general broad study. However, studying herbs can benefit an herbalist in becoming a professional and master of herbs.

Reading Herbal Recipes

- Read and study the herb.
- Don't take herb identity for granted. Not all 'organic' herbs are organic or safe.
- Stick with the recommended dose.
- Know your allergies.
- Be aware of yourself, how you feel, and reactions.
- 65 and older should start with a low, diluted dose.
- If pregnant or nursing use herbs with caution or do not use herbs at all.
- Do not give healing herbs to children under 2.
- Use caution when giving to children under 12.
- Know the interactions with other drugs, and herbs.
- Think before using a new found herb or remedy.
- Be sparingly when using oils.
- Pay attention to symptoms of toxicity.
- Keep all of your physicians informed including pharmacists.
- Discontinue herbs before any surgery, including oral surgery.

Simplers' Method of Calculating Proportions

In this lesson, we will discuss common herbs that can be found not just in Kentucky, but also in our own planted gardens or at an herb store. These herbs can be prepackaged and have a variety of uses. However, the uses for each individual herb can be overwhelming. In this lesson, we will discuss some of these herbs, their common uses, how to make them, what part of the herb to use for different needs, and common recipes.

To assist you in using the Simpler Method for recipes that use parts as measurements, we have included a measuring equivalent chart for quick reference. When making a recipe, the first thing to do is to choose your pot, double boiler, glass jar, or cup. For example, if you choose an 8 oz mug and your salve recipe calls for 1 part dandelion, 1 part chamomile, 1 part comfrey, 1 part lavender, olive oil, and beeswax, that's 4 parts of herbs. Four parts of an 8 oz mug equals 2 oz, since 4 divided into 8 equals 2. Therefore, each part equals 2 oz.

The amount of olive oil you need depends on the amount of herbs you have. The oil should barely cover the top of all of the herbs combined, with a little extra added just in case the remedy gets too hot. For the beeswax, ¼ of a cup or 1 oz is normally used for 8 oz, but sometimes you need to add a little more to make the salve thicker or a little less to make it thinner.

It is recommended to use local beeswax that is rich in color, rather than store-bought beeswax that has been condensed and does not hold the necessary constituents or phytochemical properties.

If you want to make a smaller amount of salve, you can use a tablespoon as 1 part. Fill a glass measuring cup with your herbs as indicated in parts, then add your oil with a little extra, and place the mixture in a glass jar in a pot of simmering water. Simmer for at least 2 hours, being careful not to burn or over-boil the mixture, and ensuring that the cooking area is well-ventilated.

For herbs that are very aromatic, like eucalyptus, place a jar lid with punched out holes over the jar. Once the oil has turned colors, strain the oil from the herbs using muslin cloth, assisted with a rubber band on another glass canning jar.

After straining, place the oil back in the pot with the simmering water, add the beeswax, and stir until the beeswax is melted. You can test a small amount of the mixture to see the thickness of the salve. After straining, discard the fused herbal fragments.
To make the salve more potent, you can add more herbs after the first time of straining and simmer again for another hour before straining again.
To thicken the mixture, add a teaspoon of beeswax at a time and test to see if the consistency is correct for a salve. When the mixture is complete, fill your colored container with the mixture, apply an airtight lid, keep it out of sunlight in a dry area, and label it. The salve's shelf life is about 3 months. Remember that after the salve has time to harden, the color may become darker. Allow the salve to cure for at least 24 hours before using.
TIP- For allowing the salve to cool, you can place it in the freezer for 1 hour, but make sure you take it out and cure it. Some herbalists do not cure their recipes before using, but it is important for all of the ingredients to settle.
TIP- 1 pint or 16 oz of oil will normally need about 1 ½ oz of beeswax. 1 oz of oil will need ½ tsp of beeswax.

Herbal Preparation Intake Amounts

In the following pages, we will discuss how to use herbs safely. Not all herbs are safe for everyone. Each individual is different and so are their bodies. For example, some people are allergic to eucalyptus, while others love its smell. We all have unique chemical makeups and body chemistry, so when using herbal remedies, you will find that one remedy for a 100 lb. a person may be totally different for a 150 lb. person.
There is no set recommended intake amount for herbal preparations, but most prepackaged herbal remedies or supplements have dosage instructions set for a 150 lb person. However, if you weigh only 100 lbs., you need to decrease the dosage accordingly. Always check the label of all products. For children, divide the recommended dosage by their body weight. For instance, a child would only need 1/3 of the recommended dosage, unless the product is formulated for a child's dose. Make sure you understand how to read the product label correctly. Failure to do so can result in harm.
It is important to know the herb you are taking. Some prescription drugs and over-the-counter (OTC) drugs can react with herbal remedies. As previously mentioned, some prescription drugs originated from herbs. For instance, atropine, which is used to regulate heartbeat, is from belladonna leaves, and morphine, a narcotic, is from opium poppy seeds. Ephedra, which is used in diet pills, is from the ephedra plant, which can be lethal to humans.
Some herbs are harmful to humans and animals. For example, bloodroot, comfrey, pennyroyal, and mistletoe can be toxic. Please make sure you know which herb you are using or picking in the wild. I once picked jewelweed or wild impatiens without wearing gloves because I thought it was pretty and grew by the roadside everywhere. However, as soon as I got back into my house, my hand began tingling and became numb. Unfortunately, I was experiencing a reaction to the weed. I rushed to the bathroom and cleansed my hands with a soft scrub with lemon and orange.
Later that day, I saw the tulip vine, which has a similar orange trumpet as jewelweed. I have seen cows and horses graze on this vine, but little did I know they were experiencing a horrible reaction. Later that day, I noticed that the cows and horses in that field became bloated, weak, and had diarrhea. Some of them soon died within weeks. So please remember to wear gloves and know what you are picking, even by the roadside. Some say that a little trumpet vine to an animal aids in their digestion, but a lot can be deadly.

There are guidelines to making and taking herbal remedies. Just like with prescription medicine, some herbal remedies have side effects. There are dangerous herbs, not only for humans but also for animals. Here is a checklist of guidelines to follow:

1. Before you take any herb, read about it. Get to know the herb you are about to take and its possible side effects and interactions.
2. Stick with the recommended dosage and never exceed it.
3. Remember that we are all different. Our body weight, chemistry, skin color, and age make a difference, so be aware of your body's reactions to herbal remedies.
4. If you are over 65, start with a low, diluted dose and work your way up. You will be able to tell the difference from dose to dose.
5. If you are pregnant or nursing, be very cautious of the herbs you use and touch. In my opinion, it is best to avoid ALL herbs.
6. Do not give healing herbs to children under 2 years old.

Be aware of herbal interactions with prescription medicine and over-the-counter drugs. Know what you are taking. Most of today's medicine is derived from herbs. Be sure to check the medicinal values of an herb along with your prescription medicine and over-the-counter drugs. For example, buckthorn is an herb that has a laxative value; using an OTC laxative with this herb is lethal and can cause severe abdominal pain and dehydration.

1. Keep a list of the herbs you are using and notify your physician, dentist, or any other medical doctor of any herbs that you take or use on a daily basis.
2. Discontinue any and all herbs before surgery.

Proper dosage of herbal medicine is crucial when it comes to clinical practice. To determine the proper dosage, several factors should be taken into account. These factors include the cultural context in which the herb is used, federal or state regulations, the potential toxicity or strength of each herb, the relative concentration or strength of the product, the dose form (such as tincture or standardized extract), the age, strength, and needs of the individual taking the preparation, and the duration of administration.

When it comes to cultural context, different cultures use different amounts of an herb or herb preparation for a dose and over different periods. For instance, while Native Americans used to drink several cups of strong tea made by boiling herbs, most technologically developed nations today would not find this acceptable. In China, a common herb prescription contains 5 to 10 herbs, and the daily dose for each is in the range of 3 to 12 grams. Moreover, many herb tea prescriptions given for therapeutic use contain about 60 to 200 grams of dried herbs. These are to be boiled for up to an hour, and 2 to 3 cups of the strong brew are consumed over the course of a day.

Standardized extracts are commonly prescribed in tablets and capsules, as well as teas and hydroethanolic tinctures, in Germany, other European countries, and North America. Some standardized extracts are highly concentrated. For instance, Ginkgo biloba leaves are extracted to produce a 50:1 concentrate. One part of the finished extract represents the active flavonoids and terpenes from 50 parts of the leaves.

In Europe, teas and tinctures tend to be of lower concentration than in North America, and the recommended daily dose also seems to be lower. In Germany and Europe, mother tinctures made with an extract ratio of 1:10 are often favored. This means that 10 parts of the finished tincture represent most of the desirable and active constituents from only 1 part of the herb, and this is called a "mother tincture." For instance, an ounce of echinacea mother tincture made at this concentration would represent 1/10 of an ounce of dried echinacea root or leaves, or about 3 grams. If the recommended dose were 1-2 mL, 3 times/day, the patient would receive the equivalent of about 100-200 mg of dried herb, 3 times/day, or up to 600 mg/day.

To determine the appropriate dose for each individual for any given clinical encounter, consider the strength of the herb, the strength of the preparation, the quality and freshness of the herbs that went into the product, and the size, weight, age, and needs of the patient. For instance, hydroalcoholic tinctures are usually manufactured at a concentration of anywhere from 3:1 to 10:1, and most are around 5:1. The recommended dose listed on bottles of these tincture products tends to be 20 to 80 drops, several times daily. More experienced western herbal practitioners tend to prescribe up to 5 mL of a hydroalcoholic tincture, 3 to 5 times/day, as an initial therapeutic dose, and about 1-2 mL, 2 to 3 times/day as a maintenance dose.

We recommend adjusting the dose within this range of doses based on the individual situation. Smaller people need a smaller dose than a very large person. Young children need a smaller dose than an adult. Very young children usually need only a few drops to obtain a therapeutic response. Weak or sensitive individuals need (or can tolerate) a smaller dose than a robust, healthy person. Usually, it is best to start a person on a new herb or formula at the minimum dose to check for sensitivity and response before going on to a larger dose if no response is noted. For long-term use, the dose can often be half of the therapeutic dose as a maintenance dose. It is important to note that the dose should never be thought of as static and fixed for all circumstances. Use your best judgment, taking into account the situation, and always adjust the dose rather than dispensing an herb in the same dose for every situation and person.

Potency Method (my method)

Not many herbalists practice what I call the Potency Method. This method is based on the properties or components of the chosen herbs when making a remedy for a specific purpose. The potency of a remedy is just as important as knowing which herbs complement each other and which ones counteract each other if combined in a remedy. It's important to follow a few steps before understanding the potency method: 1) know the components of the herbs, 2) know which herbs can and cannot be combined, 3) know the potency levels for each individual, such as a child, 4) know how to make the remedy, 5) create a recipe from scratch, 6) record data from proven cases and tests of success, and 7) process the remedy for sale.

To begin, you need to know the herb's components. You should have studied this in the Introduction to Herbalism lesson. You may use the reference given in that lesson, along with the chart above and the PDF chart of Healing Herbs, to help you learn about an herb's properties or components. First, you need to ask and answer a few questions about the remedy you need, for example, if you are making a first aid salve for burns, cuts, scrapes, etc.

What type of herbs do you need? For example, you may need vulnerary, anti-inflammatory, and antiseptic herbs. As a general rule, it's best to choose no more than three herbs to start with.

Can you name some herbs we could use?
Examples:

- Calendula
- Comfrey
- Aloe
- And many more...

According to the reference from the lesson Introduction to Herbalism and the Healing Herbs PDF chart, calendula and comfrey are both listed as vulneraries, which are healing agents. Therefore, to avoid potential adverse effects, it is recommended to only pick one of those herbs, calendula with aloe, or comfrey with aloe.
If you are making a remedy for a child, the elderly, or pregnant or nursing women, it is important to consider the safety of the herbs used. Calendula and comfrey are not considered safe for internal use in these groups. Therefore, it is recommended to use calendula and aloe for external use only.
To determine the potency levels of the herbs, it is important to refer to the chart provided in the document. Calendula and comfrey have stronger components than aloe, so more of those herbs should be added to the recipe.
A recommended recipe for a 1x potency salve includes:

- 2 tsp of calendula
- 1 tsp of aloe gel or oil
- 1 cup of extra virgin olive oil
- 1 oz of beeswax

It is recommended to follow the basic recipe for making the salve, and to label and cure the salve for 24 hours before testing. To test the recipe, 5-10 volunteers should try the recipe for 1 week and keep a provings journal. If the recipe proves successful for at least 7 out of 10 volunteers, there is no need to make any changes to the recipe.
To make a cold processed herbal infused oil, it is recommended to use the same herbs as before, calendula and aloe, and a 1x potency recipe. A recommended recipe includes:

- 6 tbsp of calendula
- 2 tbsp of aloe gel
- 1 cup of carrier oil

After placing the herbs in a glass jar with the carrier oil, secure with an unbleached muslin cloth or un-dyed paper towel, and sit in the window seal for 1 week. Shake daily, then strain and place the oil into the salve.
It is important to note that internal formulary is different than external, and administering home remedies to infants and young children is not recommended without professional guidance. It is recommended to follow the suggested intake consumption plan for internal remedies, and to consider safety guidelines when making any remedy.

Keeping A Journal

Keeping a journal, or what I call a database, of your experiments, clients, recipes, and provings is a great way to analyze the success of your research. Please remember to keep clients' information confidential by using numbers as a reference to their identity. Provings are the results from volunteers using a new recipe.

Database like Google & all it has to offer in documentation is a great way to save and record client information. However, it may take some time to get the program up and running, so having a paper backup file is a good idea to begin with. Remember to have a backup copy on an external hard drive or on the web. I lost my entire database when my computer crashed, even though I had the paper copies. It will take a very long time to replace 20 years worth of data.

Recording in your journal or database should become a habit. When creating a recipe, the proper records and descriptions of the making process are important. I can't tell you how many times I have forgotten to write down an ingredient when creating a recipe. Later on, when I wanted to make the recipe again, I left out an important ingredient. Before attempting to make any recipe, please read and study the herb formula before using. Know what you are using. Follow all safety guidelines not only dealing with herbs, but also prepare the kitchen area to be sanitary.

I have created the worksheets that follow for you to begin proving your recipe. I suggest starting with one recipe using the sheets provided and creating a binder as you go. Don't get overwhelmed if you cannot complete all of the information at this time. You can always go back and record it later. If you refer back to your recipe while making the remedy, be sure to use sheet protectors over your records in your binder. Once you have my samples down, feel free to create or have someone create record sheets just for you. However, you are welcome to continue printing and using mine, but please do not sell them or the template.

You can also use the worksheets provided to research the success of several different methods of the same recipe. For example, hot infused herbal oil or cold infused herbal oil. Which method works better for which ailment? Keeping this information is very important if you intend to sell the products. The scientific method is the best method to use for any experiment.

Problem
Hypothesis
Experiment
Observation
Conclusion

I have included an example of my herb journal and a blank herb journal for you to use. The blank one can be used as a guideline to go by later on in the course.

GSP & FDA

The Good Standard Process (GSP) is a quality management system that the US FDA utilizes to ensure that products are consistently produced and controlled according to established quality standards. This process is vital in regulating and monitoring the safety and efficacy of products sold to consumers. Although the GSP does not regulate herbs and plants, it has regulations regarding labeling and dispensing. These regulations aim to ensure that consumers are fully informed about the herbs and plants they purchase and that they are dispensed safely and appropriately. By adhering to these regulations, the GSP plays a critical role in protecting the health and well-being of consumers. It is important to note that one of the key regulations in place is regarding labeling. Manufacturers must adhere to labeling requirements to ensure that consumers are fully informed about the herbs and plants they are purchasing. The regulations regarding the label include details such as the product name, dosage information, expiration date, and any potential side effects. By having these details on the label, consumers can make informed decisions about the products they are purchasing, ultimately leading to better health outcomes.

I have included the GSP documentation for you as an Ebook.

The FDA's role in regulating the food industry includes ensuring that food is safe for consumption, accurately labeled, and free from harmful contaminants. They also monitor and regulate food additives, such as preservatives and coloring agents, to ensure that they do not pose any health risks to consumers.

In addition to food, the FDA also plays a crucial role in regulating the production and distribution of drugs in the United States. They ensure that drugs are safe and effective before they are approved for sale to the public. The FDA also regulates the labeling and packaging of drugs to ensure that consumers have access to accurate information about the drugs they are taking.

Overall, the FDA's role is to protect public health by ensuring that the food and drug industry is held to the highest safety standards.

As herbalists, it is crucial that we have a comprehensive understanding of all the regulations, stage guidelines, local guidelines, and laws that are relevant to our practice of herbalist consulting. Being knowledgeable about these aspects of our profession allows us to provide the best possible service to our clients, while also ensuring that we are in compliance with all relevant requirements at all times. This includes being aware of any changes or updates to these guidelines over time, as well as staying up-to-date on any new laws or regulations that may be introduced. By staying informed and always striving to improve our knowledge of these critical aspects of our work, we can continue to grow as herbalists and provide the highest level of service to our clients.

How to Study Herbs

Studying every individual herb could take a lifetime, as there are countless varieties out there. However, it's important to note that each herb has its own unique properties and uses. It's up to the herbalist to decide which herbs to study and use in their practice. Some herbalists prefer to use common backyard herbs that can be wild-harvested, while others prefer more convenient methods, such as purchasing from a company, co-op, or farmer's market. Additionally, some herbalists specialize in a particular herb for specific health reasons or symptoms, rather than for a general, broad study.

Regardless of the method chosen for studying herbs, the benefits are numerous. By understanding the properties and uses of each herb, an herbalist can create formulas tailored to the unique needs of each individual. It's important to consider not only the benefits, but also the potential side effects and drug interactions that each herb may have. By doing so, an herbalist can create safe and effective formulas for their clients.

This document provides a comprehensive guide for herbalists looking to expand their knowledge and create effective formulas for their clients. We will explore the benefits, uses, properties, side effects, warnings, possible drug interactions, and combinations with other herbs that can be used to create a formula. With this information, herbalists can become true masters of herbs and help their clients achieve optimal health and wellness.

Herb Mapping is a form of studying to map out the phytonutrients, constituents, uses of, taxonomy, etc. of each plant. Below is an example.

This allows you to fill in and record information about the herb that you choose to remember. It even helps when formulating herbal preparations and recipes. It sort of is like a thought process that is individualized just for your studies by making it your own.

Monographs

Monographs of plants are detailed written studies of a specific plant species or group of species, often including information on their taxonomy, distribution, morphology, ecology, and uses. They are typically written by botanists or other experts in the field, and are meant to provide a comprehensive understanding of the chosen plant or group of plants.

Monographs are important tools for researchers, students, and enthusiasts alike, as they provide a wealth of information that can be used to better understand and appreciate the natural world. They often include detailed descriptions of the plant's physical characteristics, such as its leaves, stem, and flowers, as well as information on its habitat and distribution.

In addition to providing information on the biology and ecology of the plant, monographs may also include details on its traditional uses in medicine, agriculture, or other industries. This can be particularly valuable for researchers looking to develop new medicines or products based on natural compounds found in plants.

Overall, monographs are an essential resource for anyone interested in the study of plants, and can provide a wealth of information on a wide range of species.

Quick Easy Sheets

I have a passion for herbal preparations and find myself creating Quick Easy Sheets and Herbal Wheels to make the process even more enjoyable.

Quick Easy Sheets are a favorite of mine because they are quick to create, simple, and provide a straightforward guide on how to use a particular plant. When making herbal preparations, it's important to have a quick reference on hand, and Quick Easy Sheets provide just that. They are perfect for beginners and experienced herbalists alike.

In addition to Quick Easy Sheets, I also create Herbal Wheels, and Infographs. These individual wheels are a unique way to organize information on specific herbs.


Manzana
Sencha
Mandarin
Lavanda
KITCHEN
Herbalism





Herbs have been treasured for their flavorful and medicinal qualities, enhancing the taste and aroma of our meals. Available in both fresh and dried forms, herbs can enrich dishes in delectable and healthy ways. Mastering their use is essential for culinary excellence. Dive into the world of cooking with herbs to learn about their characteristics, advantages, and ideal applications.

Most Popular Culinary Herbs

Herbs are a vital element of cooking, available throughout the year in fresh and dried varieties. Here are some of the most commonly used herbs in kitchens around the globe:

- Basil: With its sweet, fragrant aroma, basil is a perfect match for Mediterranean dishes, particularly in tomato-based recipes like bruschetta.
- Rosemary: The strong, pine-like flavor of rosemary pairs beautifully with poultry and grilled vegetables, adding a Mediterranean touch to meals.
- Thyme: This herb's subtle mix of earthy and citrus notes enhances a wide range of dishes, from hearty stews to delicate seafood.
- Parsley: Known for its crisp taste, parsley not only boosts flavor but also adds a pop of color as a garnish.
- Cilantro: A favorite in many international cuisines, cilantro's zesty flavor brightens up dishes such as salsa and curry.
- Mint: Celebrated for its refreshing taste, mint's cooling properties complement both sweet and savory dishes.
- Dill: Dill's delicate, feathery leaves bring subtle yet distinctive flavors, elevating sauces and pickled items.
- Oregano: A staple in Mediterranean cooking, oregano’s strong, peppery flavor enhances pizzas and pasta dishes.
- Bay Laurel: Often overlooked, bay leaves provide a gentle, aromatic flavor to slow-cooked meals like soups and sauces.
- Sage: With its earthy aroma and slightly bitter taste, sage adds unique flavor to savory recipes, especially in stuffing and risotto.

Benefits of Adding Herbs to Meals

Incorporating herbs into your daily meals—from breakfast to dinner, snacks, and drinks—offers a host of benefits, including:

- Enhanced flavor: Herbs introduce complex tastes, transforming simple meals into culinary delights.
- Nutritional boost: Packed with antioxidants, vitamins, and minerals, herbs provide numerous health benefits, such as reducing inflammation and supporting heart health.
- Digestive aid: Some herbs, like basil and oregano, not only enhance flavor but also support digestion and gut health.
- Versatility: Herbs can be used in a wide array of dishes, from appetizers to desserts, appealing to various tastes and cooking styles.

Proper storage methods for herbs can heighten their benefits while ensuring food safety.

How to Use Fresh vs. Dried Herbs

Did you know?

As a general guideline, use about one-third the quantity of dried herbs when substituting for fresh herbs in recipes.

While adding herbs to your dishes is usually straightforward, knowing the differences between fresh and dried herbs can help maximize their flavor and aroma in your cooking.

Fresh Herbs

Fresh herbs like cilantro, parsley, and dill provide an explosion of flavor and aroma. They are best added at the end of cooking or used as garnishes to maintain their vibrant qualities.

- Pros: Bright flavors, vivid colors, aromatic, ideal for quick-cooking recipes
- Cons: Short shelf life, can wilt quickly, seasonal availability

Dried Herbs

Dried herbs often have a more concentrated flavor and aroma, meaning you'll use less compared to fresh herbs. They are particularly effective in slow-cooked dishes like stews and soups, allowing their flavors to develop over time. To release their flavor, add dried herbs early in the cooking process.

- Pros: Longer shelf life, intense flavor, convenient and widely available
- Cons: Less vibrant color, may lose potency if not stored properly

Tips for Adding Herbs to Everyday Meals

With their variety of flavors and aromas, herbs can challenge even experienced cooks. The following tips can help you integrate them more easily into your cooking:

- Fresh garnishes: Just before serving, sprinkle freshly chopped parsley or cilantro on your dish to maintain their flavor and freshness in presentation.
- Infused oils and vinegars: Steep fresh herbs like rosemary, thyme, or basil in olive oil or vinegar for an added flavor boost in dressings and marinades.
- Herb butter: Combine finely chopped fresh herbs into softened butter for a tasty topping on grilled meats, roasted vegetables, or bread.
- Herb pesto: Beyond basil, try using other herbs like cilantro, mint, or parsley for a versatile pesto sauce.
- Herb-infused water or tea: Add fresh herbs like mint or basil to water and chill in the fridge for a refreshing drink.
- Herb salad dressings: Blend fresh herbs with olive oil, vinegar, mustard, and honey for a delightful dressing that enhances salads.

Exploring different herbs can enrich your culinary skills, introducing exciting flavor profiles to your dishes. Whether sourced from your thriving herb garden or your pantry's dried collection, meals infused with herbs highlight a diverse range of flavors and culinary creativity, ensuring a satisfying dining experience with every bite.

Herbs, Spices, & Alliums

Herbs, Spices, and Alliums are essential ingredients in many recipes around the world. They are used to add flavor, aroma, and color to dishes, making them more appetizing and enjoyable to eat. Each of these categories- herbs, spices, and alliums- has a distinct flavor and is used in different ways in cooking.

When it comes to herbs, they come from the leafy green part of the plant. Some of the most common herbs include basil, oregano, thyme, and parsley. These are often used in Italian, French, and Mediterranean cuisine, but they can also be found in other types of cooking.

Spices, on the other hand, come from a part of the plant other than the leaf. This can include the seed, root, or bark. Some of the most popular spices include cinnamon, cumin, ginger, and turmeric. Spices are often used in Indian, Middle Eastern, and African cuisine, but they can also be found in other types of cooking.

Finally, alliums are a family of plants that include garlic, chives, onions, and leeks. These ingredients are used to add a pungent, savory flavor to dishes. Garlic is especially popular in Italian and Mediterranean cuisine, while onions are used in a wide variety of dishes around the world.

In summary, herbs, spices, and alliums are essential ingredients in cooking that add flavor, aroma, and color to dishes. While each category has its own distinct flavor, they can all be used to enhance the taste of food and create memorable dining experiences.

Benefits of Herbs & Spices

Herbs and spices have been used for centuries for their various benefits and uses. In addition to adding flavor and increasing enjoyment of food, they have many other benefits as well. For example, herbs and spices impart characteristic flavors such as sour, sweet, salty, and tangy which can enhance the taste of food. Some herbs and spices can also act as natural tenderizers for meats, making them a useful addition to any kitchen.

Furthermore, many herbs and spices offer health benefits through their abundant phytonutrients and other physiologic effects. For instance, some herbs and spices can act on the digestive system and help with issues such as indigestion or bloating. They can also help to boost the immune system and prevent illnesses.

In conclusion, the benefits of herbs and spices are numerous and varied. They are not only a great way to add flavor to food, but they can also be used to improve health and wellness. It is important to incorporate a variety of herbs and spices into your diet to fully experience the benefits they have to offer.

Phytonutrients

Phytonutrients are an essential class of protective compounds found in plants that play a vital role in human health. These compounds are present in a wide range of plant-based foods, including herbs, spices, alliums, fruits, and vegetables. They are believed to have health-promoting properties, such as antioxidant and anti-inflammatory effects, which can help reduce the risk of chronic diseases.

Studies have shown that phytonutrients are more beneficial when consumed in whole food and plant form, such as herbs and spices, as opposed to isolated supplements. Whole foods and plants are rich in a variety of phytonutrients that work synergistically to provide optimal health benefits.

Phytonutrients are responsible for giving plants their vibrant colors, which makes them easy to spot in fruits and vegetables. However, they are also present in white foods, such as garlic, onions, and leeks, which are often overlooked as sources of phytonutrients. Consuming a variety of colorful fruits and vegetables is important to obtain a wide range of phytonutrients, which cannot be obtained solely from fruits and vegetables.

In conclusion, phytonutrients are essential compounds that play a crucial role in human health. It is important to consume a variety of plant-based foods in order to obtain the full range of health benefits associated with phytonutrients.

Herbs, Spices & Phytonutrients

This document is a comprehensive list of various compounds found in foods and supplements that are believed to have health benefits. These compounds are known as phytonutrients and can be found in a wide variety of plant-based foods, including fruits, vegetables, herbs, and spices. Phytonutrients are beneficial for health because they have antioxidant and anti-inflammatory properties, which can help reduce the risk of chronic diseases such as cancer, cardiovascular disease, and diabetes.

Below is a list of some of the most common phytonutrients and their benefits:

- Anthocyanins: Found in berries, cherries, and grapes, these compounds are believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of heart disease, cancer, and cognitive decline.
- Flavanols: Found in cocoa, tea, and wine, these compounds are believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of heart disease, stroke, and cognitive decline.
- Flavones: Found in parsley, thyme, and celery, these compounds are believed to have anti-inflammatory properties that may help reduce the risk of heart disease, cancer, and diabetes.
- Isoflavones: Found in soybeans, these compounds are believed to have estrogen-like effects that may help reduce the risk of breast cancer and osteoporosis.
- Rosemarinic acid: Found in rosemary and sage, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and Alzheimer's disease.
- Kaempferol: Found in kale, broccoli, and green tea, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and heart disease.
- Resveratrol: Found in red wine, grapes, and peanuts, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of heart disease, cancer, and Alzheimer's disease.
- Apigenin: Found in parsley and celery, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Carbazole alkaloid: Found in curry leaves, this compound is believed to have antioxidant properties that may help reduce the risk of cancer.
- Zeaxanthin: Found in egg yolks and green vegetables, this compound is believed to have antioxidant properties that may help reduce the risk of age-related macular degeneration.
- Capsaicin: Found in chili peppers, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Curcumin: Found in turmeric, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and Alzheimer's disease.

- Gingerol: Found in ginger, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Daidzein: Found in soybeans, this compound is believed to have estrogen-like effects that may help reduce the risk of breast cancer and osteoporosis.
- Genistein: Found in soybeans, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and heart disease.
- Lycopene: Found in tomatoes, this compound is believed to have antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Cinnamic acid: Found in cinnamon, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Piperine: Found in black pepper, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and Alzheimer's disease.
- Catechins: Found in green tea, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Ellagic acid: Found in berries, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Sulforaphane: Found in broccoli and other cruciferous vegetables, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Thymoquinone: Found in black seed oil, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer.
- Cyanidin: Found in berries, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Carotenoids: Found in carrots, sweet potatoes, and other orange and yellow vegetables, these compounds are believed to have antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Anethole: Found in anise, fennel, and licorice, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Myricetin: Found in berries, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Diosgenin: Found in yams, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Silybin: Found in milk thistle, this compound is believed to have antioxidant properties that may help reduce the risk of liver damage.
- Lignans: Found in flaxseed, sesame seeds, and whole grains, these compounds are believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.
- Eugenol: Found in cloves and cinnamon, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.

- Coumarin: Found in cinnamon and other spices, this compound is believed to have anti-inflammatory and antioxidant properties that may help reduce the risk of cancer and cardiovascular disease.

Incorporating a variety of these phytonutrients into your diet may help you achieve optimal health and reduce your risk of chronic disease.

Fresh or Dried?

When it comes to Kitchen Herbalism, there are a variety of factors that can influence whether fresh, dried, or both forms of herbs are best to use. While dried herbs can be more convenient to store and use year-round, fresh herbs may provide a stronger flavor and may contain more of the active compounds that are beneficial to health. Additionally, some herbs may be more effective when used in combination with other herbs, either in fresh or dried form. It's also important to consider the seasonality of herbs - certain herbs may be more readily available in fresh form during specific times of the year, while others may be more commonly found dried. Ultimately, the choice between fresh, dried, or both forms of herbs may depend on the specific needs and preferences of the individual using them, as well as the desired outcome of the recipe or remedy being created.

Storage

It's important to keep your herbs and spices fresh and readily available for optimal flavor in your dishes. To do so, store them in airtight tin or glass containers, which can be used for making fresh spice mixes or drying herbs from the garden. Keep them in a cool, dark place at 50-60°F to protect them from heat and moisture.

Herbs and spices are an essential part of cooking. They can add a depth of flavor, aroma, and taste to any dish. However, if they are not stored properly, they can lose heir potency and flavor over time.

Ground spices tend to lose their flavor more quickly than whole spices because they have a larger surface area exposed to air. This can cause them to become stale and lose their aroma. To maintain their flavor, it's best to use ground spices within six months, although some experts recommend using them within one year.

Whole spices, on the other hand, can last up to two years if stored properly. They are more resistant to oxidation and can retain their flavor and aroma for longer periods.

In addition to their flavor benefits, herbs and spices also have health benefits. They are rich in antioxidants and anti-inflammatory compounds that can help boost your immune system and reduce inflammation in the body.

In summary, keep your herbs and spices fresh by storing them in airtight containers, in a cool, dark place at 50-60°F. Use ground spices within six months and whole spices within two years. Conduct a nose test to check the freshness of your spices, and enjoy the many flavor and health benefits they provide!

Keep herbs and spices readily available: • Store them in airtight tin or glass containers, which can be used for making fresh spice mixes or drying herbs from the garden. • Store them in a cool, dark place at 50-60°F. • Some experts recommend using ground spices within 6 months, while others say they can last up to 1 year. Whole spices can last up to 2 years. • Use the nose test - open the bottle and if there is little aroma, toss it out. If it's a whole spice, rub it between your fingers - it should release volatile oil that can be felt and smelled.